Juan David Arroyo Lopez

Civil Engineer | Geotechnics & Tunnelling

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Summary

Civil Engineer (Honors) from Universidad Nacional de Colombia with 2+ years of experience in geotechnical research specializing in both physical-mechanical soil characterization and mathematical modeling, and over 6 months of experience in tunnel design and consultancy. As a DAAD alumni, I completed a year-long international exchange at the Karlsruhe Institute of Technology (KIT) in Germany, developing advanced technical skills and high adaptability. I am proficient in geotechnical and structural design software, Geographic Information Systems (GIS), and programming languages, with a proven track record of commitment and responsibility in international research and high-impact engineering projects.

Education

Universidad Nacional de Colombia, BSc in Civil Engineering – Bogotá D.C., Colombia Mar 2020 – Apr 2026
Focus on Geotechnics, Structures, and Modeling.

- Graduated with Honors.
- GPA: 4.7/5.0 (Colombian scale, 0-5; minimum passing grade 3.0).

Karlsruhe Institute of Technology, Exchange (without a degree) in Civil Engineering – Karlsruhe, Germany Oct 2024 – Sept 2025
Academic exchange with emphasis on Geotechnics, Geological Engineering, Hydrogeology, and data-driven modeling with Python.

Experience

Civil Engineer I, AECOM – Bogotá D.C., Colombia Apr 2026 – present
Tunnelling and Trenchless

- Supported the design and technical review of cavern components for the Ontario Line South (OLS) and the Ontario Pumped Storage (OPS) projects.
- Performed geotechnical analyses, ground-behaviour assessments, and damage assessments for buildings and utilities using PLAXIS 2D/3D, RS2, RS3, and Python.
- Prepared technical reports, drawings, and calculation notes for client and internal review, in line with AASHTO, ACI, CSA, and FHWA standards.
- Developed Python tools that automated calculation and data processing, reducing processing time by 80%.

Engineering Assistant, HINCCO SAS – Gualmatán, Colombia Jan 2026 – Apr 2026
Participation in a 500m urban road paving project in Gualmatán, Nariño (Colombia).

- Supervised day-to-day construction activities to ensure compliance with technical specifications.
- Produced daily site diaries and project documentation.
- Coordinated subcontractors and labor crews to keep the project on schedule.
- Tracked material deliveries, inventory, and resource allocation.
- Interpreted blueprints and designs for the construction team to ensure precise execution.

Intern, Institut für Bodenmechanik und Felsmechanik (IBF) – Karlsruhe, Germany Apr 2025 – Aug 2025

Research internship as part of the KOSPIE 24/25 scholarship.

- Optimized laboratory workflows by integrating robotic automation in geotechnical testing.
- Developed a Python-based tool to automate soil sample preparation using a Cobot (collaborative robot).
- Authored technical user manuals for students and laboratory technicians to standardize automated processes.

Engineering Assistant, Soil Laboratory at Universidad Nacional de Colombia – Bogotá D.C., Colombia Sept 2022 – Aug 2024

- Collaborated in the preparation and structuring of research proposals for civil engineering, geotechnics, and pavement projects.
- Conducted Geotechnical Design Reports for 7 government buildings.
- Performed technical evaluations for the construction, reinforcement, and restoration of institutional infrastructure.
- Researched geotechnical applications for biosolids incineration ash, focusing on sustainable construction materials.
- Executed field and laboratory tests for advanced geotechnical and structural analysis.

Publications

Robotic automated specimen preparation (RASP) technique for soil mechanics June 2026

Hans Henning Stutz, Luis Mugele, Jochen Zürn, *Juan David Arroyo Lopez*
10.53243/ICSMGE2026-268 (Proceedings of the 21st International Conference on Soil Mechanics and Geotechnical Engineering (ICSMGE), Vienna, Austria)

Robotic automated specimen preparation (RASP): applied to air pluviation June 2026

Luis Mugele, *Juan David Arroyo Lopez*, Jochen Zürn, Hans Henning Stutz
10.1007/s11440-026-03112-4 (Acta Geotechnica)

Awards

Combined Study and Practice Stays for Engineers from Developing Countries (KOSPIE) Aug 2024 – July 2025

DAAD scholarship for academic exchange and internship in Germany.

- Selected through open call at the Faculty of Engineering in Bogotá, based on academic performance and proficiency in English and German.

Skills

Languages: Spanish (native), English (IELTS Academic 7.5, C1), German (B2, TestDaF).

Software: Python, PLAXIS2D, PLAXIS3D, RS2, RS3, Slide2, QGIS, AutoCAD, Civil 3D, SAP2000.

Soft Skills: Proactivity, Effective Communication, Adaptability, Teamwork, Organization, Problem-Solving.

Certificates

PLAXIS 3D: Getting Started Oct 2025

Course 3: Numerical Hydrogeological Modeling Applied to Geothermal Energy Sept 2026

Consulting Bootcamp Oct 2025

Lexium Cobot - EcoStruxure Cobot Expert Configuration May 2025

Personal Interests and Hobbies

Interests: Science, technology, programming, languages, travel, cultures

Hobbies: Cooking, hiking, listening to music